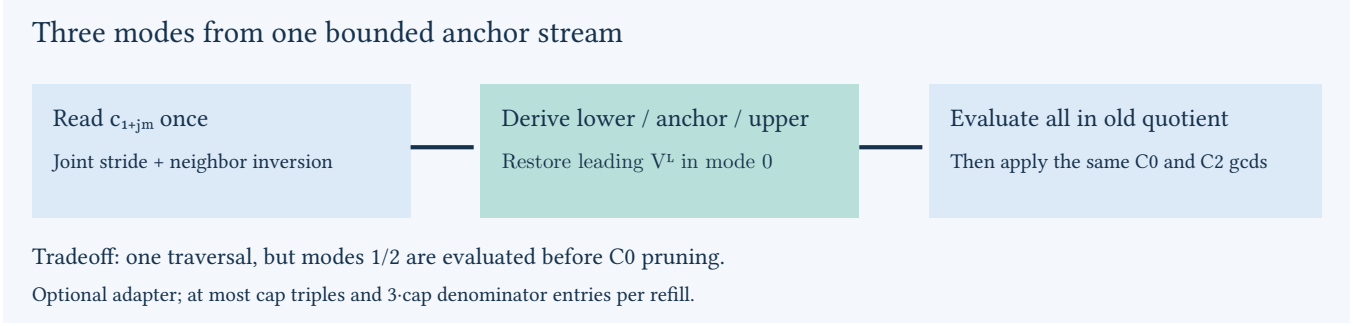
Triple-mode coefficient reuse

T-1893 · Prime-field direct-image update · 2026-11-08

The optional update generates three selected modes from one mode1 anchor stream. Adjacent-binomial identities provide both neighbors; their denominators join the stride-transition batch inverse. Mode0′s known leading coefficient is restored separately.



Construction. For h=(r−1)/2, m≥3 dividing h, and n=1+jm, derive $c_{n-1} = -n\frac{c_n}{h-n+1}$ and $c_{n+1} = -(h-n)\frac{c_n}{n+1}$. Both denominators are nonzero. The reader holds at most cap triples and 3·cap denominator entries; no full coefficient vector is retained.

Quotient reuse. Compute F0,F1,F2 modulo the old branch polynomial g. After g1=gcd(g,C0), these representatives remain valid because g1 divides g. Applying the same nonlinear C2 gcd therefore preserves the staged support. Source schedule, initialization and final point verification remain unchanged.

Exact qualification. validation01 passes 40 reader configurations, 975 coefficient equalities, 1,920 quotient equalities and 3,640 candidate-state equalities. The optional adapter matches outcomes, routes and source calls for 843 public target/cap cases; 735 subsequent updates exercise the new path. Inherited stdout is byte-identical. PCC-20261108-130f40 is unsigned producer evidence; signed review pending.

Cost tradeoff. If a branch survives C0, the staged update performs two anchor traversals; the fused update performs one. But it adds neighbor work, a third component and a leading power, and evaluates modes1/2 before pruning. If C0 kills every branch, staged evaluation avoids that quotient work. These correctness checks do not select a faster policy.

Evidence and next. Sources, reversible generation, compiler hashes, raw logs and derivation are retained. No cloud/timed or production comparison ran. Next freeze a matched fused/staged update comparison, including zero-survivor cases and full quotient/coefficient accounting. Original cohort and separate transfer research remain intact.